

Clockwork

An AI operations platform for trades businesses — designed and built by Ian McCallum

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Work leaks out of a trades business at ten different points in a day: the call that rang out while someone was under a sink, the quote they meant to send, the follow-up they forgot, the appointment nobody confirmed, the invoice still in the glovebox. Clockwork closes every one of them. An AI agent drafts the work — the reply, the quote, the reminder, the invoice — and the owner approves it. I designed and built the whole system as sole architect and engineer.

Ten automated stages. Eight screens. One rule: the agent drafts, the owner sends.

WHAT IT AUTOMATES

01	Catch the missed call	Texts back in seconds, before the customer calls the next shop.
02	Brief the lead	Marks it hot, warm, or cold and names the likely parts and hours from the symptoms alone.
03	Draft the reply	Reads the thread and writes the owner's answer in the owner's voice.
04	Price the quote	Computed from the rate card and sent as a link the customer accepts from their phone.
05	Chase the follow-up	On a schedule until they answer — and it stops the moment they do.
06	Confirm the appointment	So a truck never rolls to an empty house.
07	Collect the invoice	Pay-by-text on a hosted page; the job flips to paid on its own.
08	Ask for the review	Every won job requests one automatically.
09	Mine the book	Surfaces quiet customers and aging equipment, ranked by what they're likely worth.
10	Plan tomorrow	Written overnight against the board; the owner approves each move in the morning.

ENGINEERING DECISIONS

- **Prices are computed, never generated.** Quotes come from the owner's own rate card rather than model output, so the system cannot hallucinate a number that costs somebody money. Deterministic logic and the language model are deliberately kept on separate sides of the line.
- **Earned autonomy, not assumed autonomy.** Every message starts as a draft for the owner to approve. The agent only graduates to sending a given message type once enough drafts have passed unedited — a deliberate two-step switch, revocable in one tap.
- **Ten voice rules, enforced in CI.** The agent's writing is governed by a written specification, and the test suite fails the build on any violation — so message quality is a gate, not a hope.
- **A failure is never recorded as an answer.** An unreachable model produces a distinct, retryable state rather than a silently degraded reply filed as finished work. If a figure can't be computed, the screen says so instead of showing a zero.
- **Every send is on the record.** Nothing goes out without an outbox row carrying a timestamp and a reason, and opt-outs are enforced at every send path rather than checked in one place and assumed everywhere else.
- **Multi-tenant from day one.** The same system serves a single owner-operator and a multi-location business, with role-separated access for owners, office staff, and techs, and strict data isolation between locations.

Status: built and demo-ready, with founding pricing open to the first 25 Illinois contractors.

ALSO BUILT

Imanol VillaGomez

imanolvillagomez.com

Portfolio site for a concert videographer whose work covers Charli XCX, Playboi Carti, and The Hellp.

Nics Marketing

nicsmarketing.co

Agency site with a React prequalification tool that screens client readiness before intake.

ianmccallum.com

ianmccallum.com

Personal site built from scratch as an interactive desktop-style interface.

TMM Photography

tmmphotos.com

Responsive site for a sports photographer with galleries and session booking.

Fortivus Academy

fortivusrugby.com

Brand-driven site for a rugby training academy, aimed at athletes and their families.

beatyourclock.com

beatyourclock.com

Marketing site for Beat the Clock, with an interactive product simulation and a free time-loss assessment.

Python

Prompt Engineering

LLM Integration

AI Agent Design

Multi-Tenant Architecture

Messaging & Payments

JavaScript

React